

Recurrent Neural Networks (RNNs)

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Lecture 13

ANN & DL

What is a Neural Network?

- A neural network is a system inspired by the human brain
- It consists of:
 - Input layer
 - Hidden layers
 - Output layer
- It can learn patterns from data

Limitation of Traditional Neural Networks

- Traditional Neural Networks assume independent inputs.
- Cannot handle:
 - Time-series data
 - Sequential patterns
- Example:
 - Sentence: “He went to the bank”
- → Meaning depends on previous words

What is Sequential Data?

- Data where order matters
- Examples:
 - Text (NLP)
 - Speech signals
 - Stock prices
 - Weather data

What is an RNN?

- A Recurrent Neural Network (RNN) is a neural network designed for sequential data.
- It has memory of previous inputs.
- The key Idea is that the output depends on
 - Current input
 - Previous hidden state

Why Do We Need RNN?

- To model Language, Time-series and Sequences data.
- Captures temporal dependencies and key features.
- Real-Life Examples
 - Google Translate
 - Speech recognition
 - Chatbots
 - Predicting next word in a sentence
 - Spellcheck

RNN vs CNN

Feature	CNN	RNN
Data Type	Images	Sequential
Memory	No	Yes
Order Awareness	No	Yes

→ CNN Captures spatial features for example **identifies words**

→ RNN Captures temporal dépendances and maintains context (**understands sentence meaning**)

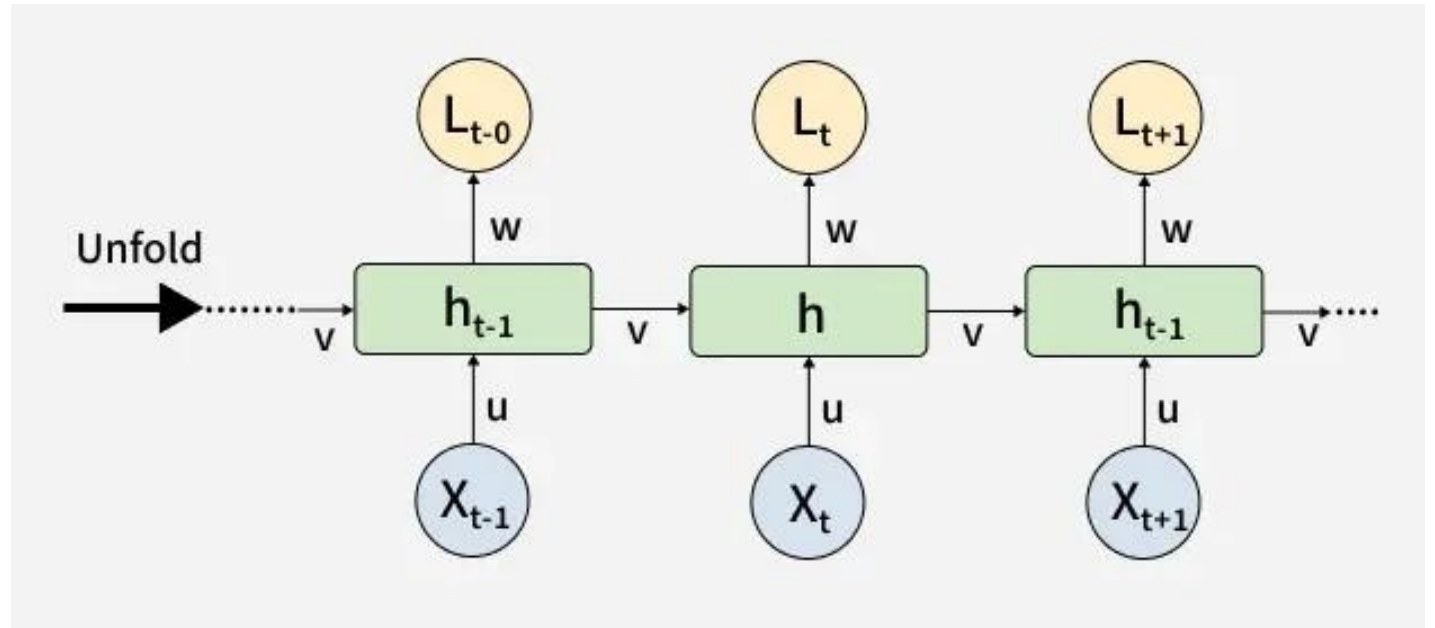
RNN ARCHITECTURE

At time step t :

Input: x_t

Hidden state: h_t

Output: y_t



RNN Formula

Hidden state:

$$h_t = f(W_h h_{t-1} + W_x x_t + b)$$

Output:

$$y_t = W_y h_t$$

Where

- x_t : input at time t
- h_t : hidden state (memory)
- W : weights
- b : bias
- f : activation function (tanh/ReLU)

Types of RNN Architectures

1. One-to-One (binary classification spam or not spam)
2. One-to-Many (Image captioning)
3. Many-to-One (Sentiment analysis)
4. Many-to-Many (Translation)

Activation Functions

- $\tanh$ → commonly used in RNN
- sigmoid → used in gating
- ReLU → is very less common in RNN

Q1: A simple RNN processes a sequence of 2-time steps
Inputs: $(x_1 = 1, x_2 = 2)$,
Weights: $(W_x = 1, W_h = 1)$, Bias: $(b = 0)$, Initial hidden state $h_0 = 0$. Find: h_1, h_2

Solution: Formula $h_t = W_h h_{t-1} + W_x x_t$

Step 1 (t = 1)

$$h_1 = (1)(0) + (1)(1) = 1$$

$$h_1 = 1$$

Step 2 (t = 2)

$$h_2 = (1)(1) + (1)(2) = 3$$

$$h_2 = 3$$

Final Answer

Hidden states: $h_1 = 1$ and $h_2 = 3$

Interpretation

RNN accumulates information over time
Output depends on previous state

Q2. An RNN processes sequence over time steps. If Inputs are ($x_1 = 2, x_2 = 3$), Weights: ($W_x = 2, W_h = 1$), Bias: ($b = 0$), Initial hidden state $h_0 = 0$. Find: h_1, h_2

Solution:

Formula $h_t = W_h h_{t-1} + W_x x_t$

Step 1 (t = 1)

Step 2 (t = 2)



Thank you